

Section 4.3: Measures of Dispersion

Finite Mathematics MTH 107 Fall 2014

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Mars Hill University

Friday 31st October, 2014



/home/jim/JamesAshe/MHU/images/MarsHillU

Dispersion

Munson	185	135	200	185	250	155
McCracken	182	185	188	185	180	190

Dispersion

Munson	185	135	200	185	250	155
McCracken	182	185	188	185	180	190

135 155 185 185 200 250

Dispersion

Munson	185	135	200	185	250	155
McCracken	182	185	188	185	180	190

Munson

135 155 185 185 200 250

$\bar{x}$ Mean:

$$\frac{135+155+185+185+200+250}{6} =$$

185

Dispersion

Munson	185	135	200	185	250	155
McCracken	182	185	188	185	180	190

Munson

135 155 185 185 200 250

$\bar{x}$ Mean:

$$\frac{135+155+185+185+200+250}{6} =$$

185

$$\text{Median: } \frac{185 + 185}{2} = 185$$

Dispersion

Munson	185	135	200	185	250	155
McCracken	182	185	188	185	180	190

Munson

135 155 185 185 200 250

$\bar{x}$ Mean:

$$\frac{135+155+185+185+200+250}{6} =$$

185

$$\text{Median: } \frac{185 + 185}{2} = 185$$

Mode: 185

McCracken

Dispersion

Munson	185	135	200	185	250	155
McCracken	182	185	188	185	180	190

Munson

135 155 185 185 200 250

$\bar{x}$ Mean:

$$\frac{135+155+185+185+200+250}{6} =$$

185

$$\text{Median: } \frac{185 + 185}{2} = 185$$

Mode: 185

McCracken

180 182 185 185 188 190

Dispersion

Munson	185	135	200	185	250	155
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Munson

135 155 185 185 200 250

$\bar{x}$ Mean:

$$\frac{135+155+185+185+200+250}{6} =$$

185

$$\text{Median: } \frac{185 + 185}{2} = 185$$

Mode: 185

McCracken

180 182 185 185 188 190

$\bar{x}$ Mean:

$$\frac{180+182+185+185+188+190}{6} =$$

185

Dispersion

Munson	185	135	200	185	250	155
McCracken	182	185	188	185	180	190

Munson

135 155 185 185 200 250

$\bar{x}$ Mean:

$$\frac{135+155+185+185+200+250}{6} =$$

185

$$\text{Median: } \frac{185 + 185}{2} = 185$$

Mode: 185

McCracken

180 182 185 185 188 190

$\bar{x}$ Mean:

$$\frac{180+182+185+185+188+190}{6} =$$

185

$$\text{Median: } \frac{185 + 185}{2} = 185$$

Dispersion

Munson	185	135	200	185	250	155
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Munson

135 155 185 185 200 250

$\bar{x}$ Mean:

$$\frac{135+155+185+185+200+250}{6} =$$

185

$$\text{Median: } \frac{185 + 185}{2} = 185$$

Mode: 185

McCracken

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$\bar{x}$ Mean:

$$\frac{180+182+185+185+188+190}{6} =$$

185

$$\text{Median: } \frac{185 + 185}{2} = 185$$

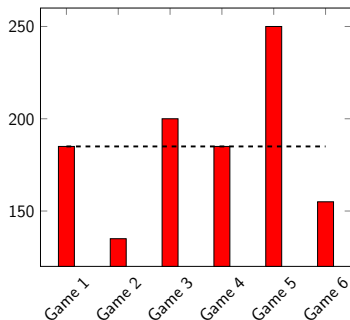
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Dispersion

Munson	185	135	200	185	250	155
McCracken	182	185	188	185	180	190

Dispersion

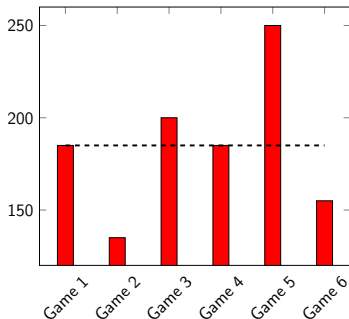
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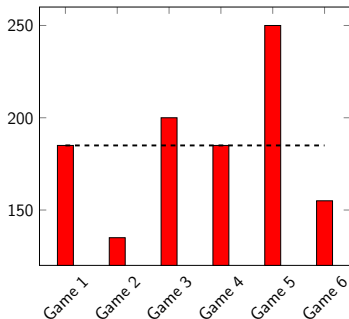
Munson



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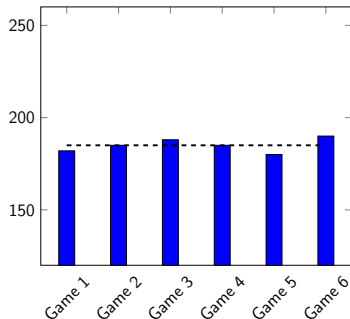
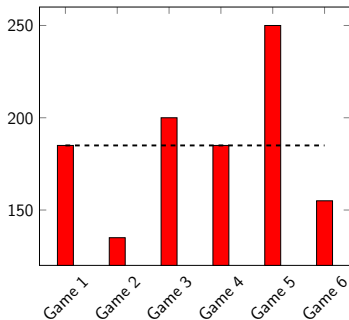
Munson



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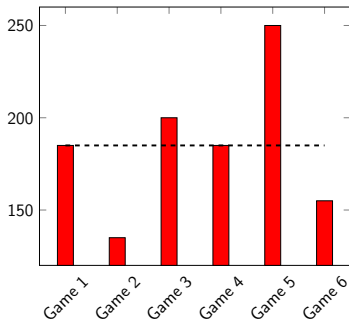
Munson



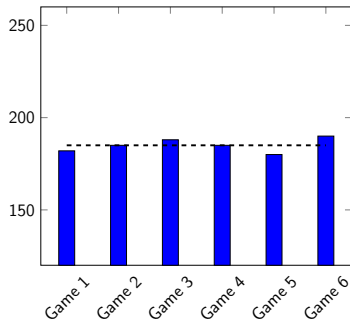
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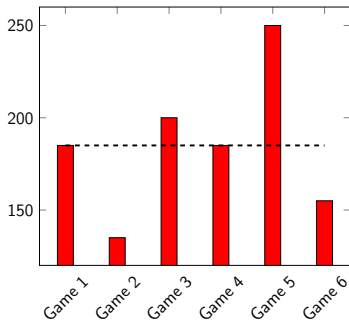
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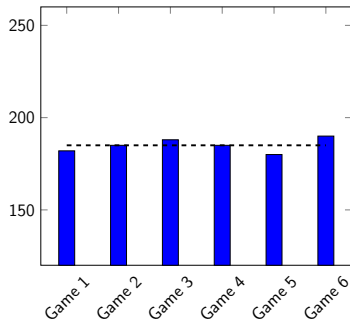
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McCracken



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Deviation: $x - \bar{x}$	0	-50	15	0	65	-30
McCracken	182	185	188	185	180	190
Deviation: $x - \bar{x}$	-3	0	3	0	-5	5

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The “average” deviation

Dispersion

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Munson

The “average” deviation

$$\frac{0-50+15+0+65-30}{6}$$

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Munson

The “average” deviation

$$\frac{0-50+15+0+65-30}{6}$$
$$= \frac{0}{6} = 0$$

Dispersion

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McCracken

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Munson

The “average” deviation

$$\frac{0-50+15+0+65-30}{6}$$
$$= \frac{0}{6} = 0$$

McCracken

The “average” deviation

$$\frac{-3+0+3+0-5+5}{6}$$

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Munson

The “average” deviation

$$\frac{0-50+15+0+65-30}{6}$$
$$= \frac{0}{6} = 0$$

NOT the measurement of dispersion we need.

McCracken

The “average” deviation

$$\frac{-3+0+3+0-5+5}{6}$$
$$= \frac{0}{6} = 0$$

Dispersion

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Munson

The “average” deviation

$$\frac{0-50+15+0+65-30}{6}$$
$$= \frac{0}{6} = 0$$

NOT the measurement of dispersion we need.

(This always happens!)

McCracken

The “average” deviation

$$\frac{-3+0+3+0-5+5}{6}$$
$$= \frac{0}{6} = 0$$

Variance and Standard Deviation

Dispersion

Variance and Standard Deviation

Dispersion

Sample Variance Deviation Given
a sample of n data points, x_1 ,
 $x_2, \dots, x_n$, the **variance**, s^2 , of
the data is

$$s^2 = \frac{\sum (x - \bar{x})^2}{n - 1}$$

Sample Variance Deviation Given a sample of n data points, $x_1, x_2, \dots, x_n$, the **variance**, s^2 , of the data is

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Variance

Data is almost always a sample.

Population will vary more than a sample, so $n - 1$ is used so the variance will not be underestimated.

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Bessel's correction

Variance and Standard Deviation

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Standard Deviation

Given a sample of n data points, $x_1, x_2, \dots, x_n$, the **standard deviation**, s , of the data is

$$s = \sqrt{\text{variance}}$$

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Population will vary more than a sample, so $n - 1$ is used so the variance will not be underestimated.

Bessel's correction

Standard Deviation

The “average” distance from the mean.

Variance and Standard Deviation

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$x - \bar{x}$	0	-50	15	0	65	-30

Variance and Standard Deviation

Munson	185	135	200	185	250	155
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$(x - \bar{x})^2$						

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$(x - \bar{x})^2$	0^2	$(-50)^2$				

Variance and Standard Deviation

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Variance and Standard Deviation

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Variance and Standard Deviation

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$x - \bar{x}$	0	-50	15	0	65	-30
$(x - \bar{x})^2$	0 ²	(-50) ²	15 ²	0 ²	65 ²	(-30) ²
	0	2500				

Variance and Standard Deviation

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	0	2500	225	0	4225	900

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Sum: $0 + 2500 + 225 + 0 + 4225 + 900 = 7850$

Variance and Standard Deviation

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$$s^2 = \frac{7850}{n - 1}$$

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Sum: $0 + 2500 + 225 + 0 + 4225 + 900 = 7850$

$$s^2 = \frac{7850}{n - 1} = \frac{7850}{5} = 1570$$

Variance and Standard Deviation

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$$s^2 = \frac{7850}{n - 1} = \frac{7850}{5} = 1570$$

$$s = \sqrt{1570}$$

Variance and Standard Deviation

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	0	2500	225	0	4225	900

Sum: $0 + 2500 + 225 + 0 + 4225 + 900 = 7850$

$$s^2 = \frac{7850}{n - 1} = \frac{7850}{5} = 1570$$

$$s = \sqrt{1570} \approx 39.623\dots$$

Variance and Standard Deviation

Munson	185	135	200	185	250	155
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$$s = \sqrt{1570} \approx 39.623 \dots$$

McCracken	182	185	188	185	180	190
$x - \bar{x}$	-3	0	3	0	-5	5

Variance and Standard DeviationDispersion

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$(x - \bar{x})^2$	$(-3)^2$	0^2	3^2	0^2	$(-5)^2$	

Variance and Standard Deviation

Munson	185	135	200	185	250	155
---------------	-----	-----	-----	-----	-----	-----

$$s = \sqrt{1570} \approx 39.623 \dots$$

McCracken	182	185	188	185	180	190
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	9	0				

Variance and Standard Deviation

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	9	0	9	0	25	25

Variance and Standard Deviation

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$$\text{Sum: } 9 + 0 + 9 + 0 + 25 + 25 = 68$$

Variance and Standard Deviation

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	9	0	9	0	25	25

$$\text{Sum: } 9 + 0 + 9 + 0 + 25 + 25 = 68$$

$$s^2 = \frac{68}{n-1}$$

Variance and Standard Deviation

Munson	185	135	200	185	250	155
---------------	-----	-----	-----	-----	-----	-----

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	9	0	9	0	25	25

$$\text{Sum: } 9 + 0 + 9 + 0 + 25 + 25 = 68$$

$$s^2 = \frac{68}{n-1} = \frac{68}{5} = 13.6$$

Variance and Standard Deviation

Munson	185	135	200	185	250	155
---------------	-----	-----	-----	-----	-----	-----

$$s = \sqrt{1570} \approx 39.623 \dots$$

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$x - \bar{x}$	-3	0	3	0	-5	5
$(x - \bar{x})^2$	$(-3)^2$	0^2	3^2	0^2	$(-5)^2$	5^2
	9	0	9	0	25	25

$$\text{Sum: } 9 + 0 + 9 + 0 + 25 + 25 = 68$$

$$s^2 = \frac{68}{n-1} = \frac{68}{5} = 13.6$$

$$s = \sqrt{13.6}$$

Variance and Standard Deviation

Munson	185	135	200	185	250	155
---------------	-----	-----	-----	-----	-----	-----

$$s = \sqrt{1570} \approx 39.623 \dots$$

McCracken	182	185	188	185	180	190
$x - \bar{x}$	-3	0	3	0	-5	5
$(x - \bar{x})^2$	$(-3)^2$	0^2	3^2	0^2	$(-5)^2$	5^2
	9	0	9	0	25	25

$$\text{Sum: } 9 + 0 + 9 + 0 + 25 + 25 = 68$$

$$s^2 = \frac{68}{n-1} = \frac{68}{5} = 13.6$$

$$s = \sqrt{13.6} \approx 3.687 \dots$$

Standard Deviation of Grouped Data

Grade		Freq.		
$50 \leq x < 60$		3		
$60 \leq x < 70$		13		
$70 \leq x < 80$		68		
$80 \leq x < 90$		13		
$90 \leq x < 100$		3		
		$n = 100$		

Standard Deviation of Grouped Data

Grade	Midpoint	Freq.		
$50 \leq x < 60$		3		
$60 \leq x < 70$		13		
$70 \leq x < 80$		68		
$80 \leq x < 90$		13		
$90 \leq x < 100$		3		
		$n = 100$		

Standard Deviation of Grouped Data

Grade	Midpoint	Freq.		
$50 \leq x < 60$	55	3		
$60 \leq x < 70$	65	13		
$70 \leq x < 80$	75	68		
$80 \leq x < 90$	85	13		
$90 \leq x < 100$	95	3		
		$n = 100$		

Alt. Formula for Variances² =

$$\frac{1}{n-1} \left(\sum x^2 - \frac{(\sum x)^2}{n} \right)$$

Standard Deviation of Grouped Data

Grade	Midpoint	Freq.	$f \cdot x$	
$50 \leq x < 60$	55	3		
$60 \leq x < 70$	65	13		
$70 \leq x < 80$	75	68		
$80 \leq x < 90$	85	13		
$90 \leq x < 100$	95	3		
$n = 100$				

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$50 \leq x < 60$	55	3	165	
$60 \leq x < 70$	65	13		
$70 \leq x < 80$	75	68		
$80 \leq x < 90$	85	13		
$90 \leq x < 100$	95	3		
$n = 100$				

Alt. Formula for Variances² =

$$\frac{1}{n-1} \left(\sum x^2 - \frac{(\sum x)^2}{n} \right)$$

Standard Deviation of Grouped Data

Grade	Midpoint	Freq.	$f \cdot x$	
$50 \leq x < 60$	55	3	165	
$60 \leq x < 70$	65	13	845	
$70 \leq x < 80$	75	68	5100	
$80 \leq x < 90$	85	13	1105	
$90 \leq x < 100$	95	3	285	
		$n = 100$		

Alt. Formula for Variances² =

$$\frac{1}{n-1} \left(\sum x^2 - \frac{(\sum x)^2}{n} \right)$$

Standard Deviation of Grouped Data

Grade	Midpoint	Freq.	$f \cdot x$	
$50 \leq x < 60$	55	3	165	
$60 \leq x < 70$	65	13	845	
$70 \leq x < 80$	75	68	5100	
$80 \leq x < 90$	85	13	1105	
$90 \leq x < 100$	95	3	285	
		$n = 100$	$\Sigma f \cdot x =$	
			7500	

Alt. Formula for Variances² =

$$\frac{1}{n-1} \left(\Sigma x^2 - \frac{(\Sigma x)^2}{n} \right)$$

Standard Deviation of Grouped Data

Grade	Midpoint	Freq.	$f \cdot x$	$f \cdot x^2$
$50 \leq x < 60$	55	3	165	
$60 \leq x < 70$	65	13	845	
$70 \leq x < 80$	75	68	5100	
$80 \leq x < 90$	85	13	1105	
$90 \leq x < 100$	95	3	285	
		$n = 100$	$\Sigma f \cdot x =$	
			7500	

Alt. Formula for Variances² =

$$\frac{1}{n-1} \left(\Sigma x^2 - \frac{(\Sigma x)^2}{n} \right)$$

Standard Deviation of Grouped Data

Grade	Midpoint	Freq.	$f \cdot x$	$f \cdot x^2$
$50 \leq x < 60$	55	3	165	9075
$60 \leq x < 70$	65	13	845	
$70 \leq x < 80$	75	68	5100	
$80 \leq x < 90$	85	13	1105	
$90 \leq x < 100$	95	3	285	
		$n = 100$	$\Sigma f \cdot x =$	
			7500	

Alt. Formula for Variances² =

$$\frac{1}{n-1} \left(\Sigma x^2 - \frac{(\Sigma x)^2}{n} \right)$$

Standard Deviation of Grouped Data

Grade	Midpoint	Freq.	$f \cdot x$	$f \cdot x^2$
$50 \leq x < 60$	55	3	165	9075
$60 \leq x < 70$	65	13	845	54925
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$90 \leq x < 100$	95	3	285	
		$n = 100$	$\sum f \cdot x =$	
			7500	

Alt. Formula for Variances² =

$$\frac{1}{n-1} \left(\sum x^2 - \frac{(\sum x)^2}{n} \right)$$

Standard Deviation of Grouped Data

Grade	Midpoint	Freq.	$f \cdot x$	$f \cdot x^2$
$50 \leq x < 60$	55	3	165	9075
$60 \leq x < 70$	65	13	845	54925
$70 \leq x < 80$	75	68	5100	382500
$80 \leq x < 90$	85	13	1105	93925
$90 \leq x < 100$	95	3	285	27075
		$n = 100$	$\Sigma f \cdot x =$	
			7500	

Alt. Formula for Variances² =

$$\frac{1}{n-1} \left(\Sigma x^2 - \frac{(\Sigma x)^2}{n} \right)$$

Standard Deviation of Grouped Data

Grade	Midpoint	Freq.	$f \cdot x$	$f \cdot x^2$
$50 \leq x < 60$	55	3	165	9075
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$80 \leq x < 90$	85	13	1105	93925
$90 \leq x < 100$	95	3	285	27075
		$n = 100$	$\Sigma f \cdot x =$	$\Sigma f \cdot x^2 =$
			7500	567500

Alt. Formula for Variances² =

$$\frac{1}{n-1} \left(\Sigma x^2 - \frac{(\Sigma x)^2}{n} \right)$$

Standard Deviation of Grouped Data

Grade	Midpoint	Freq.	$f \cdot x$	$f \cdot x^2$
$50 \leq x < 60$	55	3	165	9075
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$80 \leq x < 90$	85	13	1105	93925
$90 \leq x < 100$	95	3	285	27075
		$n = 100$	$\sum f \cdot x =$	$\sum f \cdot x^2 =$
			7500	567500

Alt. Formula for Variance:

Grouped data $s^2 =$

$$\frac{1}{n-1} \left(\sum f \cdot x^2 - \frac{(\sum f \cdot x)^2}{n} \right)$$

Standard Deviation of Grouped Data

Grade	Midpoint	Freq.	$f \cdot x$	$f \cdot x^2$
$50 \leq x < 60$	55	3	165	9075
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$90 \leq x < 100$	95	3	285	27075
		$n = 100$	$\sum f \cdot x =$	$\sum f \cdot x^2 =$
			7500	567500

Alt. Formula for Variance:

Grouped data $s^2 =$

$$\frac{1}{n-1} \left(\sum f \cdot x^2 - \frac{(\sum f \cdot x)^2}{n} \right)$$

$$s^2 = \frac{1}{99} \left(567500 - \frac{(7500)^2}{100} \right) \\ \approx 50.5051$$

Standard Deviation of Grouped Data

Grade	Midpoint	Freq.	$f \cdot x$	$f \cdot x^2$
$50 \leq x < 60$	55	3	165	9075
$60 \leq x < 70$	65	13	845	54925
$70 \leq x < 80$	75	68	5100	382500
$80 \leq x < 90$	85	13	1105	93925
$90 \leq x < 100$	95	3	285	27075
		$n = 100$	$\sum f \cdot x =$	$\sum f \cdot x^2 =$
			7500	567500

Alt. Formula for Variance:

Grouped data $s^2 =$

$$\frac{1}{n-1} \left(\sum f \cdot x^2 - \frac{(\sum f \cdot x)^2}{n} \right)$$

$$s^2 = \frac{1}{99} \left(567500 - \frac{(7500)^2}{100} \right) \\ \approx 50.5051$$

$$s = \sqrt{50.5051} \approx 7.1067$$

Distributions

